In [ ]:
```
# EXPERIMENT 7 – HANDLING OUTLIERS & MISSING VALUES IN A DATASET
PIN NO.: 241UDS0099
## AIM
To preprocess a dataset by removing duplicates, detecting outliers, and handling missing values using Pandas and Scikit-learn SimpleImputer.
```

In [ ]:
```
## Step 1: Import the Basic Libraries
```

In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.impute import SimpleImputer
```

In [ ]:
```
## Step 2: Load & Analyze the Dataset
```

In [ ]:
```
df_241UDS0099 = pd.read_csv("Experiment_7_241UDS0099_Academic_Performance.csv")
df_241UDS0099.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 123 entries, 0 to 122
Data columns (total 24 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   Branch                  120 non-null    str
 1   Marks[10th]             120 non-null    float64
 2   Marks[12th]             119 non-null    float64
 3   Gender                  121 non-null    str
 4   Board[10th]             121 non-null    str
 5   Board[12th]             121 non-null    str
 6   Category                121 non-null    str
 7   GPA 1                   121 non-null    float64
 8   Rank                    122 non-null    float64
 9   Normalized Rank         121 non-null    float64
 10  CGPA                    119 non-null    float64
 11  Current Back            121 non-null    float64
 12  Ever Back               119 non-null    float64
 13  GPA 2                   121 non-null    float64
 14  GPA 3                   121 non-null    float64
 15  GPA 4                   121 non-null    float64
 16  GPA 5                   121 non-null    float64
 17  GPA 6                   121 non-null    float64
 18  Olympiads Qualified     121 non-null    float64
 19  Technical Projects      121 non-null    float64
 20  Tech Quiz               121 non-null    float64
 21  Engg. Coaching          121 non-null    float64
 22  NTSE Scholarships       121 non-null    float64
 23  Miscellany Tech Events  121 non-null    float64
dtypes: float64(19), str(5)
memory usage: 23.2 KB
```

In [ ]:
```
df_241UDS0099.head()
```

Out[3]: Columns 1–8 of 22

	Branch	Marks[10th]	Marks[12th]	Gender	Board[10th]	Board[12th]	Category
0	NaN	NaN	83.17	Male	BSEB Patna	BIEAP	EWS
1	CIVIL	87.75	78.02	Male	CBSE	State Board	EWS
2	EEE	96.43	94.12	Female	State Board	CBSE	OBC
3	CSE	53.00	83.88	Female	ICSE	State Board	OBC
4	MECH	73.21	73.61	Male	State Board	CBSE	SC

Columns 9–15 of 22 · first column repeated

	GPA 1	Rank	Normalized Rank	...	GPA 3	GPA 4	GPA 5
0	6.01	45403.0	0.756717	...	6.64	5.39	5.44
1	6.67	46377.0	0.772950	...	6.41	6.12	6.17
2	7.29	51804.0	0.863400	...	6.94	7.28	6.98
3	6.19	34581.0	0.576350	...	6.26	6.13	5.40
4	6.61	1844.0	0.030733	...	6.64	6.42	6.34

Columns 16–22 of 22 · first column repeated

	GPA 6	Olympiads Qualified	Technical Projects	Tech Quiz	Engg. Coaching	NTSE Scholarships	Miscellany Tech Events
0	6.11	1.0	1.0	0.0	0.0	0.0	3.0
1	6.50	1.0	3.0	1.0	1.0	1.0	1.0
2	7.25	2.0	5.0	2.0	0.0	0.0	3.0
3	5.92	1.0	1.0	2.0	2.0	1.0	2.0
4	6.81	2.0	3.0	3.0	1.0	0.0	2.0

5 rows × 24 columns

In [ ]:
```
df_241UDS0099.shape
```

Out[4]: (123, 24)

In [ ]:
```
df_241UDS0099 = df_241UDS0099.dropna(how="all")
df_241UDS0099.shape
```

Out[5]: (122, 24)

In [ ]:
```
## Step 3: Find & Remove the Duplicates
```

In [ ]:
```
df_241UDS0099[["Marks[10th]", "Marks[12th]", "CGPA"]].duplicated().sum()
```

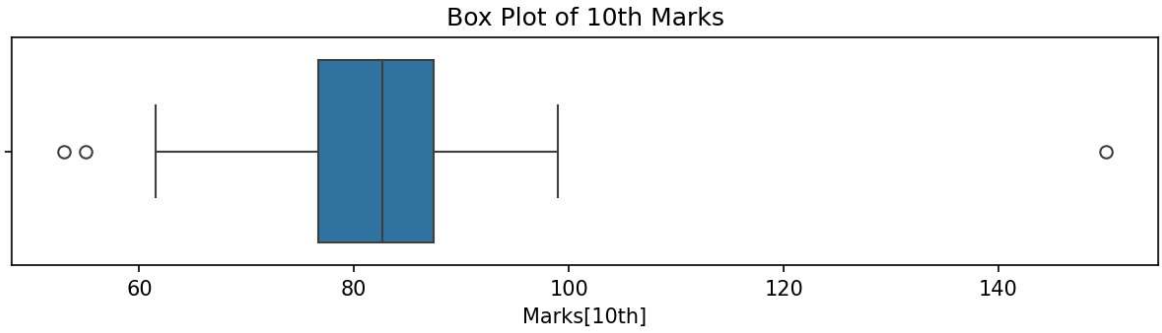
Out[6]: np.int64(2)

In [ ]:
```
df_241UDS0099 = df_241UDS0099.drop_duplicates()
df_241UDS0099.shape
```

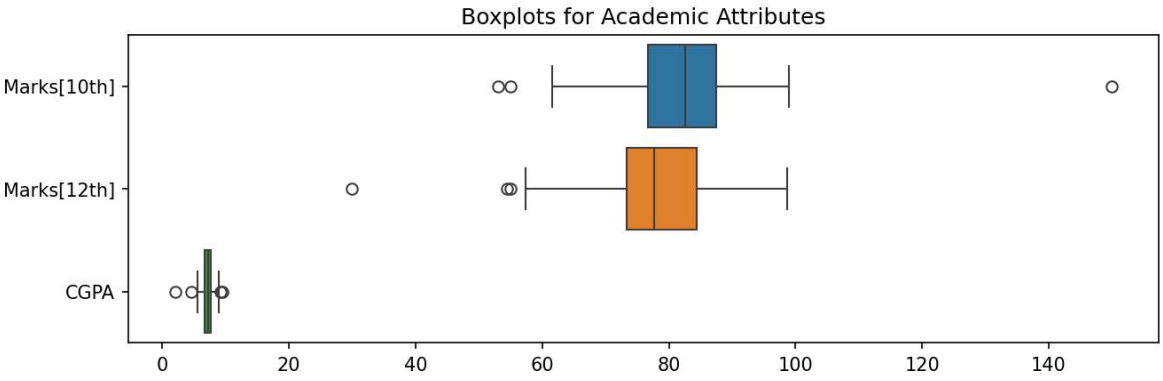
Out[7]: (120, 24)

In [ ]:
```
## Step 4: Identify & Remove the Outliers
```

In [ ]:
```
plt.figure(figsize=(10, 2), dpi=150)
sns.boxplot(data=df_241UDS0099, x="Marks[10th]")
plt.title("Box Plot of 10th Marks")
plt.show()
```



In [ ]:
```
columns_241UDS0099 = ["Marks[10th]", "Marks[12th]", "CGPA"]
plt.figure(figsize=(10, 3), dpi=150)
sns.boxplot(data=df_241UDS0099[columns_241UDS0099], orient="h")
plt.title("Boxplots for Academic Attributes")
plt.show()
```



In [ ]:
```
Q1_241UDS0099 = df_241UDS0099[columns_241UDS0099].quantile(0.25)
Q3_241UDS0099 = df_241UDS0099[columns_241UDS0099].quantile(0.75)
IQR_241UDS0099 = Q3_241UDS0099 - Q1_241UDS0099
Q1_241UDS0099, Q3_241UDS0099, IQR_241UDS0099
```

Out[10]:
```
(Marks[10th]    76.6925
 Marks[12th]    73.3400
 CGPA            6.7200
 Name: 0.25, dtype: float64,
 Marks[10th]    87.4525
 Marks[12th]    84.4200
 CGPA            7.6800
 Name: 0.75, dtype: float64,
 Marks[10th]    10.76
 Marks[12th]    11.08
 CGPA            0.96
 dtype: float64)
```

In [ ]:
```
outliers_241UDS0099 = (
    (df_241UDS0099[columns_241UDS0099] < (Q1_241UDS0099 - 1.5 * IQR_241UDS0099)) |
    (df_241UDS0099[columns_241UDS0099] > (Q3_241UDS0099 + 1.5 * IQR_241UDS0099))
)
outliers_241UDS0099.sum()
```

Out[11]:
```
Marks[10th]    3
Marks[12th]    3
CGPA           4
dtype: int64
```

In [ ]:
```
df_241UDS0099 = df_241UDS0099.loc[~outliers_241UDS0099.any(axis=1)].copy()
df_241UDS0099.shape
```

Out[12]:
```
(110, 24)
```

In [ ]:
```
## Step 5: Fill the Missing Values
```

In [ ]:
```
df_241UDS0099.isnull().sum()
```

Out[13]:
```
Branch                   2
Marks[10th]              2
Marks[12th]              3
Gender                   1
Board[10th]              1
Board[12th]              1
Category                 1
GPA 1                    1
Rank                     0
Normalized Rank          1
CGPA                     3
Current Back             1
Ever Back                2
GPA 2                    1
GPA 3                    1
GPA 4                    1
GPA 5                    1
GPA 6                    1
Olympiads Qualified      1
Technical Projects       1
Tech Quiz                1
Engg. Coaching           1
NTSE Scholarships        1
Miscellany Tech Events   1
dtype: int64
```

In [ ]:
```
df_241UDS0099["Marks[10th]"] = df_241UDS0099["Marks[10th]"].fillna(
    df_241UDS0099["Marks[10th]"].mean()
)
df_241UDS0099["Marks[10th]"].isnull().sum()
```

Out[14]:
```
np.int64(0)
```

In [ ]:
```
numerical_imputer_241UDS0099 = SimpleImputer(strategy="median")
df_241UDS0099[["Marks[12th]", "CGPA"]] = numerical_imputer_241UDS0099.fit_transform(
    df_241UDS0099[["Marks[12th]", "CGPA"]]
)
df_241UDS0099[["Marks[12th]", "CGPA"]].isnull().sum()
```

Out[15]:
```
Marks[12th]    0
CGPA           0
dtype: int64
```

In [ ]:
```
df_241UDS0099["Branch"].value_counts()
```

Out[16]:
```
Branch
CSE      31
EEE      22
ECE      18
CIVIL    13
MECH     11
PROD      7
IT        6
Name: count, dtype: int64
```

In [ ]:
```
categorical_imputer_241UDS0099 = SimpleImputer(strategy="most_frequent")
df_241UDS0099[["Branch"]] = categorical_imputer_241UDS0099.fit_transform(
    df_241UDS0099[["Branch"]]
)
df_241UDS0099["Branch"].value_counts()
```

Out[17]:
```
Branch
CSE      33
EEE      22
ECE      18
CIVIL    13
MECH     11
PROD      7
IT        6
Name: count, dtype: int64
```

In [ ]:
```
df_241UDS0099.info()
```

```
<class 'pandas.DataFrame'>
Index: 110 entries, 0 to 119
Data columns (total 24 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   Branch                  110 non-null    str
 1   Marks[10th]             110 non-null    float64
 2   Marks[12th]             110 non-null    float64
 3   Gender                  109 non-null    str
 4   Board[10th]             109 non-null    str
 5   Board[12th]             109 non-null    str
 6   Category                109 non-null    str
 7   GPA 1                   109 non-null    float64
 8   Rank                    110 non-null    float64
 9   Normalized Rank         109 non-null    float64
 10  CGPA                    110 non-null    float64
 11  Current Back            109 non-null    float64
 12  Ever Back               108 non-null    float64
 13  GPA 2                   109 non-null    float64
 14  GPA 3                   109 non-null    float64
 15  GPA 4                   109 non-null    float64
 16  GPA 5                   109 non-null    float64
 17  GPA 6                   109 non-null    float64
 18  Olympiads Qualified     109 non-null    float64
 19  Technical Projects      109 non-null    float64
 20  Tech Quiz               109 non-null    float64
 21  Engg. Coaching          109 non-null    float64
 22  NTSE Scholarships       109 non-null    float64
 23  Miscellany Tech Events  109 non-null    float64
dtypes: float64(19), str(5)
memory usage: 21.5 KB
```

In [ ]:
```
df_241UDS0099.isnull().sum()
```

Out[19]:
```
Branch                   0
Marks[10th]              0
Marks[12th]              0
Gender                   1
Board[10th]              1
Board[12th]              1
Category                 1
GPA 1                    1
Rank                     0
Normalized Rank          1
CGPA                     0
Current Back             1
Ever Back                2
GPA 2                    1
GPA 3                    1
GPA 4                    1
GPA 5                    1
GPA 6                    1
Olympiads Qualified      1
Technical Projects       1
Tech Quiz                1
Engg. Coaching           1
NTSE Scholarships        1
Miscellany Tech Events   1
dtype: int64
```

In [ ]:
```
## RESULT
Duplicate records were removed, outliers were identified using the IQR method and removed, and selected missing values were
filled using mean, median, and most-frequent strategies.
```